Exploring Heterometallic Co–Ni–Cu Benzenedicarboxylate Coordination Polymers via Automated Ternary Gradient Synthesis

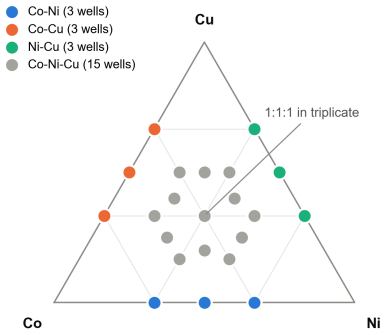
Project description | SDL Hackathon, UZH HTEL, 26–30 October 2026
Team MOFKeKing (project M3-MOF): Aritra Roy, Piyush Ranjan Maharana, Tarak Nath Das

1. Background and aim

Benzenedicarboxylate (BDC) metal–organic frameworks are studied for their tunable electronic[1] and magnetic[2] properties, and bimetallic BDC frameworks are well established[3,4]. The ternary Co–Ni–Cu–BDC system, however, has not been explored across a wide composition range. We asked whether a self-driving-lab workflow (automated dispensing plus fast UV–Vis screening) can map this space and flag compositions whose spectra cannot be explained by simpler mixes, as leads for HPLC–MS and PXRD.

2. Experimental design

24 Co:Ni:Cu compositions, all with at least two metals: the 1:1, 2:1 and 1:2 mixes on each binary edge (9) and 15 ternaries (1:1:1 in triplicate to measure noise, plus 2:1:1, 2:2:1 and 3:2:1 permutations). Each composition runs at metal : ligand 1:1, 1:2, 1:3 and 1:4, giving 96 reactions with 5.4 µmol total metal (18 mM) in 300 µL DMF.



Metal acetates (450 mM) and BDC (180 mM) stocks are diluted to 90 mM and dispensed by the OT-2, BDC last so all reactions start together. Microwave-assisted synthesis, which gives bimetallic BDC frameworks rapidly[5,6], was our preferred heating method, but no high-throughput microwave reactor was available, so the capped vials were heated in a conventional oven for 13 h at 120 °C instead.

3. What was executed and what was not

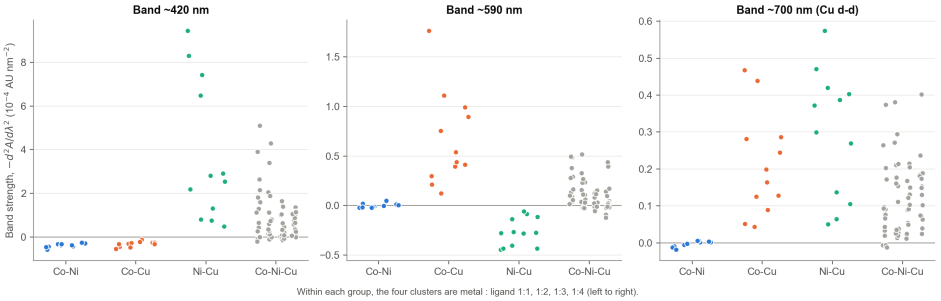
Step	Status	Notes
Design and stock calculations	Done	Python scripts and an Excel calculator.
Stock and working solutions	Done (HTEL staff)	Prepared by hand following the submitted procedure.
OT-2 dispensing into 96 × 2 mL GC vials	Done (HTEL staff)	DMF, Co, Ni, Cu, then BDC last; 420 transfers.
Reaction: 120 °C oven	Done (HTEL staff)	Oven instead of microwave (no high-throughput microwave reactor available); 13 h instead of the planned 3 h.
UV–Vis, 400–900 nm, all 96 reactions	Done	50 µL aliquots in DMF-safe glass inserts instead of the planned 200 µL; no DMF blank.
HPLC–MS	Not yet available	Pending at the time of writing.
PXRD of the solids	Not done	Planned follow-up on the leads.

4. Analysis approach

- **Solid formation.** Suspended solid scatters light and raises the whole spectrum, so the median absorbance from 860 to 900 nm ($A_{scatter}$) is used as a measure of how much solid formed.
- **Bands.** A Savitzky–Golay second derivative (40 nm window) removes the smooth scattering background and turns absorption bands into peaks. Band strength is the mean signal in windows at ~420, ~460, ~590 and ~700 nm.
- **Noise.** The 1:1:1 triplicates give the measurement noise at each wavelength (CV of $A_{scatter}$ 5–29% across the four ratios).
- **Three-metal test.** Each ternary spectrum is rebuilt from the best non-negative mix of the nine binary spectra at the same ratio. The leftover is scored as PNR (peak-to-noise ratio: biggest leftover signal divided by the noise). As a null, each binary is rebuilt from the other eight.

5. Key results

- **Copper drives solid formation.** Co–Ni mixes scatter least ($A_{scatter}$ 0.17 to 0.80); Co–Cu (0.91 to 1.63) and Ni–Cu (1.00 to 1.71) scatter strongly. More BDC gives more solid.
- **A ~700 nm band (Cu(II) d–d)** appears in every Cu-containing mix and is absent without Cu.
- **A ~590 nm band needs Co and Cu together** (0.46 to 0.77 in Co–Cu, -0.02 to 0.03 in Co–Ni with the same Co content), strongest at Co:Cu 1:1 and 2:1.
- **A ~420 nm band needs Ni and Cu together** (1.63 to 6.64 in Ni–Cu, absent elsewhere), strongest at Ni:Cu 1:1, 2:1 and 1:2 with little BDC. These two bands are **bimetallic signatures**.
- **No three-metal feature.** All 60 ternaries are rebuilt from binaries to within PNR 8.4, against a binary null of 25.1 (95th percentile). 35/60 ternaries exceed PNR = 3, but so do 21/36 binaries.



Band strength by metal pair: ~420 nm needs Ni + Cu, ~590 nm needs Co + Cu, and the ~700 nm Cu band is absent without Cu.

6. Success criteria

Criterion (application)	Outcome
1. UV–Vis spectra for all 96 reactions	**Met.** 96/96 spectra, 400–900 nm.
2. A UV–Vis feature absent from simpler compositions (PNR > 3)	**Partly met.** Two bimetallic bands; no ternary-only feature.
3. LRMS ion consistent with a mixed-metal stoichiometry	**Pending.** HPLC–MS results not yet available.
4. Ternary composition map with compositions of interest	**Met** for UV–Vis (scattering and band maps at all four ratios).

7. Next steps

- **HPLC–MS and PXRD on the leads** (Co:Cu 1:1 and 2:1; Ni:Cu 1:1, 2:1 and 1:2; 1:1:1, 3:2:1 and 2:1:3 ternaries at high BDC) to look for mixed-metal ions and to tell a new phase from a physical mixture.

References

1 Panda, J. P.; Singha, D.; Panda, P. K.; Tripathy, B. C.; Rana, M. K. Experimental and DFT Study of Transition Metal Doping in a Zn-BDC MOF to Improve Electrical and Visible Light Absorption Properties. *J. Phys. Chem. C* **2022**, 126 (30), 12348–12360.

2 Thorarinsdottir, A. E.; Harris, T. D. Metal–Organic Framework Magnets. *Chem. Rev.* **2020**, 120 (16), 8716–8789.

3 Mohamedien, H. A.; Enaiet Allah, A.; Kamal, S. M.; Mohamed, F. Two-Dimensional Fe-MOF and Bimetallic NiFe-MOFs with Different Ni : Fe Ratios for Superior Electrochemical Performance in Supercapacitor Applications. *Nanoscale Adv.* **2026**, 8 (3), 896–911.

4 Saad, M.; Elhamad, A.; Deeb, M.; Alrakkad, H. From Plastic Bottle Waste to Functional Adsorbent: A Nanostructured Bimetallic Al/Cd-BDC MOF for Enhanced Methylene Blue Removal. *Sci. Rep.* **2026** (just accepted).

5 Kurisingal, J. F.; Babu, R.; Kim, S.-H.; Li, Y. X.; Chang, J.-S.; Cho, S. J.; Park, D.-W. Microwave-Induced Synthesis of a Bimetallic Charge-Transfer Metal Organic Framework: A Promising Host for the Chemical Fixation of CO_2. *Catal. Sci. Technol.* **2018**, 8 (2), 591–600.

6 Katugampalage, T. R.; Waribam, P.; Ratanatawanate, C.; Shahid, M.; Opaprakasit, P.; Chooaksorn, W.; Sreearunothai, P. Rapid Microwave-Assisted Synthesis of Fe/Co Bimetallic Metal-Organic Frameworks and Evaluating the Role of Coordinatively Unsaturated Sites for Selective CO_2 Capture. *Trends Sci.* **2025**, 22 (10), 10552.